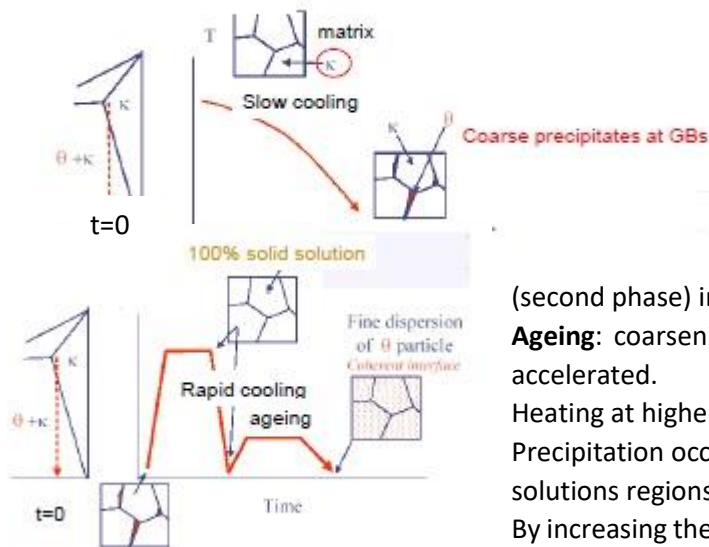


Precipitation and Aging heat treatment (H. T.): effect of cooling rate

Al-Cu alloy (96% Al-4%Cu)



Al-4%Cu alloy

Solutionizing H.T.: heating the alloy to above solvus temperature and holding for a sufficient time

Quenching: highly metastable structure will promote nucleation of precipitates inside grains

Precipitation: growth of intermetallic phases (second phase) inside $k (= \alpha)$ grains.

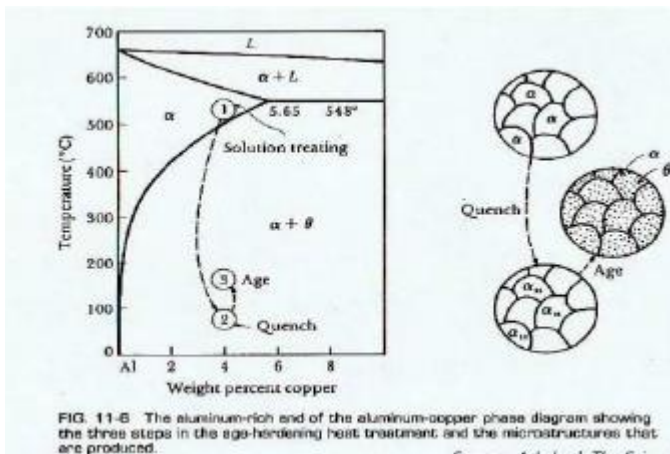
Ageing: coarsening of precipitates. As T increases ageing will be accelerated.

Heating at higher $T \rightarrow$ Rapid cooling $\rightarrow$ Precipitation $\rightarrow$ Ageing
Precipitation occurs in all alloys whose phase diagrams have solid solutions regions.

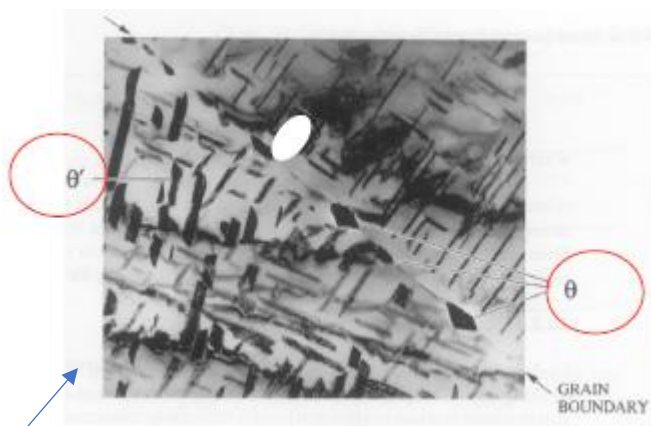
By increasing the amount of Cu and T , the amount of Al decreases.

After quenching, we can either have few large precipitates or a lot of small precipitates $\rightarrow$ **nucleation and growth**. Any phase in metals or ceramics can be grown by heating. If the phase is controlled by diffusion, it will grow according to two steps:

- 1) **Nucleation:** after a phase transformation ($\alpha \rightarrow \gamma, L \rightarrow S$), depending on the dragging force that the system is receiving from outside, there will be a number of nucleation sites into the system $\rightarrow$ solidification process which occurs at eutectic temperature (like for casting), the liquid is getting solid by quenching $\rightarrow$ we are providing energy from the outside to promote this phase transformation, from high to low T passing through the solidification T which is T_E ; because of this dragging force for the transformation, applied by the quenching, there will be same nucleation points into the liquid (Al-Cu, A + B, A and B randomly distributed at high T). During quenching, we have frozen the system, which will be in a super saturated condition $\rightarrow$ atoms after a short time will start to move and fluctuate in solid state; during fluctuation and movement at short distance, they will dynamically aggregate and whenever they aggregate in a critical number of atoms, they will form the so called "**nucleus**". When we reach critical size of nucleus, it may grow to particles and the second step, growth, will start.
- 2) **Growth:** Diffusion will contribute to expand that area by increasing the volume fraction of the new phase. If I quench very rapidly the liquid, I am putting into the system very high dragging force, therefore the transformation will occur also in a very short time. If undercooling is very large, the number of nuclei will be large and these nuclei may grow only if they reach a critical size $\rightarrow$ minimum number of atoms will aggregate and start to grow. If undercooling is not so large, the transformation will occur but the nuclei will be smaller and in this case we may have small nuclei and large phase with many small crystals. When the particles will start to expand, if we have many nuclei into the phase, whenever the nuclei touch each other they will stop growing. Therefore, very high and severe quenching will produce very fine microstructure; on the other hand, if the dragging force on the undercooling is not so large, we will have small nuclei but large phase.



longer distances → acceleration of the precipitation process (super saturation condition).



At room T, we have natural ageing.

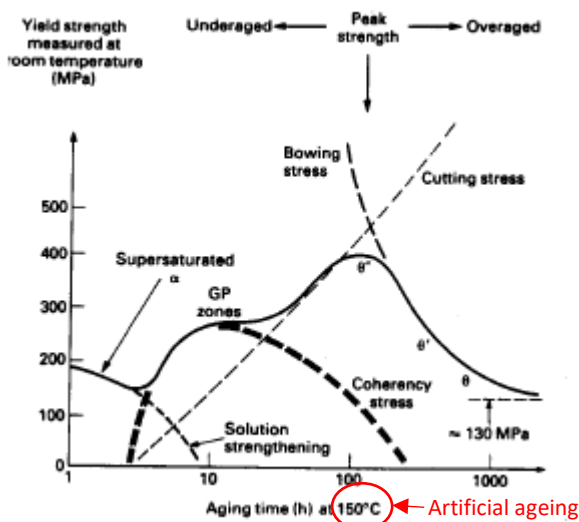
As T increases, ageing will be accelerated, and we have artificial ageing.

During slow cooling, the matrix will reject atoms in excess → these atoms will form θ at GBs.

θ' → from ageing

θ → second phase particles

Microstructure of an Al-Cu alloy with Gunier-Preston zones: these zone ensure the maximum lattice distortion, thereby promote severe strengthening of the alloy. Visible is the coherent interface along the overall length of the precipitate (15 x 150 nm).



Increasing of yield strength after ageing

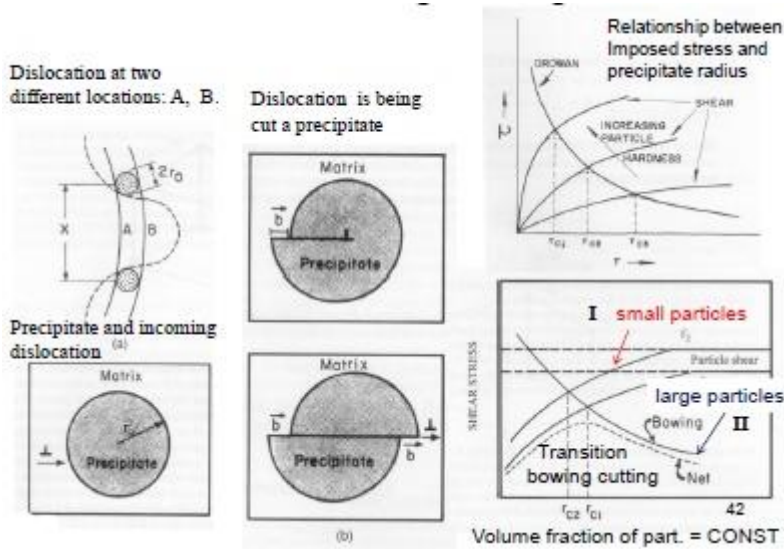
After quenching, some atoms are rearranging into solid solution and Cu atoms, which are Cu_2 , will displace from point to another looking for aggregation with Al in order to make Al-Cu compound, Al_2Cu . By diffusion, we can measure the strength which decreases after every step of diffusion. Rearrangement of diffusion → solution strengthening when atoms were frozen after quenching. The matrix could interact with the solute by means of residual stresses.

With diffusion, distortion of the lattice is recovered → returning a little bit to equilibrium by softening. With the elimination of residual stresses, stress decreases → after rearrangement caused by residual stresses (they push

atoms to move after quenching) stress will decrease a little bit, but after a certain time of rearrangement, the stress starts to increase → some precipitates are starting to originate and to grow quite fast in an early

stage. After a short time, these small precipitates will increase the strength of the alloy; after a certain time, we have a boost of precipitates → increase of strength by growth of precipitates.
 With increase of time and energy provided → strengthening translated to softening → σ_y decreases → particles are growing more and more → θ .

Orowan's strengthening model



← interaction of particles with a dislocation. The dislocation experiences bending under the pressure of a stress which is pushing the dislocation to move; the dislocation has two possibilities when meeting particles:

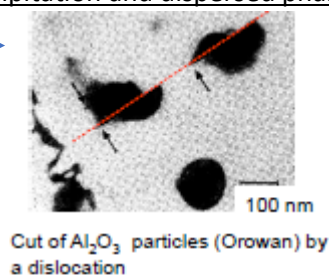
- 1) Dislocation will shear/cut particles, likely to occur when precipitates are small and coherent → dislocations can shear particles because they can move quite well into the crystal lattice of the grain but also of the particle because when they are small, they are coherent.

- 2) If the particles are too large, they are incoherent → the dislocation will be stopped and it will bow/bend → bending expresses resistance of movement of dislocations, therefore strengthening.

There are 3 steps for shearing → largest strengthening obtained when particles are neither too small nor too large because when particles are too small and we are subjecting the sample to pulling, dislocation will move and interact with particles not too large. Therefore, the dislocation will shear the particles.
 Aging → particles are growing while the volume fraction of particle is increasing.

Strengthening by precipitation and dispersed phases

Shearing of particles after a dislocation passes through

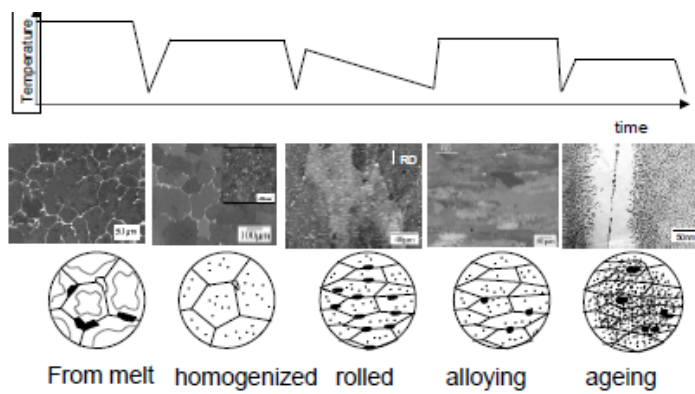


← Frank-Read → when a dislocation passes, it bows leaving a loop

$$\tau_{\text{Orowan}} \approx \frac{1}{2} \frac{Gb}{l} \sqrt{f}$$

l = distance between precipitate
 G = tangential modulus
 b = Burgers vector modulus
 f = particle volume fraction

Production of a Wrought – heat treated Al-Cu alloy



Best precipitation obtained when all second phases atoms are disgregated and the material is homogenized.

Heterogeneous structure is unsuitable for engineering applications → re-heating of solidified microstructure to make it homogeneous at high T → atoms spread into volume of the material.

During plastic deformation (rolling), the material will be shaped and its thickness will be reduced. The grain will elongate

and while T decreases, precipitates will also occur at gbs and within grain.

Alloying reduces precipitates which enhance brittleness (they will disbond the grains) → Precipitates are dissolved by making them finer especially at gbs. We don't lose the plastic deformation because grains are still long.

Precipitation Strengthening in Al alloys (7xxx)

Material	Yield Strength ksi	%elongation
Pure annealed Al	2.5	60
Solid solution Strengthened with 1% Mn	6	45
Highly cold worked pure Al	22	15
Precipitation hardened alloy 7075	80	10

← Schematization of what the strengthening mechanism is in case of a very high strength aluminum alloys used in aerospace applications.

Strengthening by twins/stacking faults

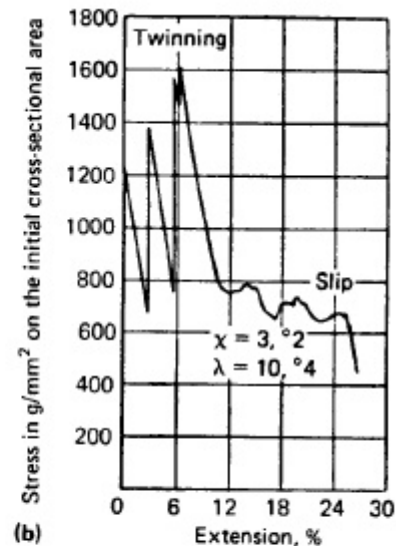
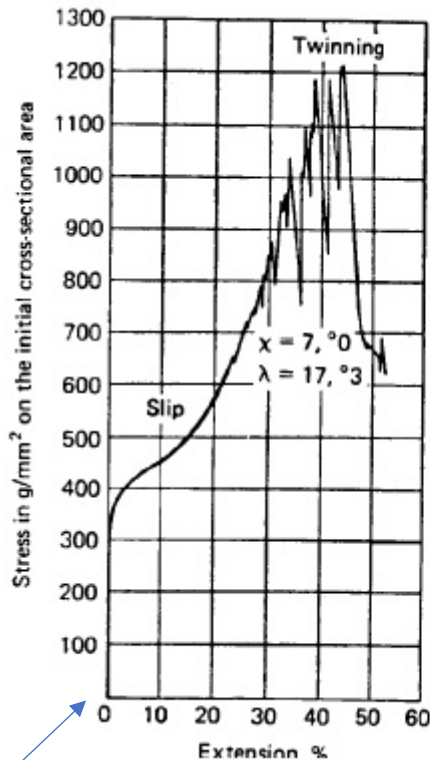
Mostly inside the grain, small amount around the boundary.

The particles inside the grain are responsible of strengthening because they interact with the dislocation movement and any slowing down of the dislocation is translated into strengthening. They contribute indirectly to strengthening by avoiding grain growth during heat treatment.

Plastic deformation by slips and twinning

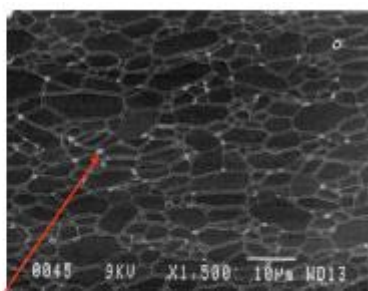
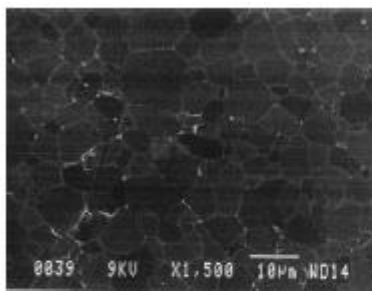
As discussed before, twinning is promoted by:

- Triaxial stress, as the material is constrained and cannot displace freely → any plastic deformation will occur through twinning
- Hexagonal crystals which can strain quite easily on the base plane like FCC crystals
- High speed deformation loading at very low T.



Stress – Strain curves different from the ones seen for ductile materials → twinning: after we reach the σ_y (around 300MPa), we have some disturbances, oscillations of the diagrams (manifestation of twinning) → material likely to strain by twinning

5) Strengthening by hard dispersed second phases



GBs particle strengthen the alloy by :

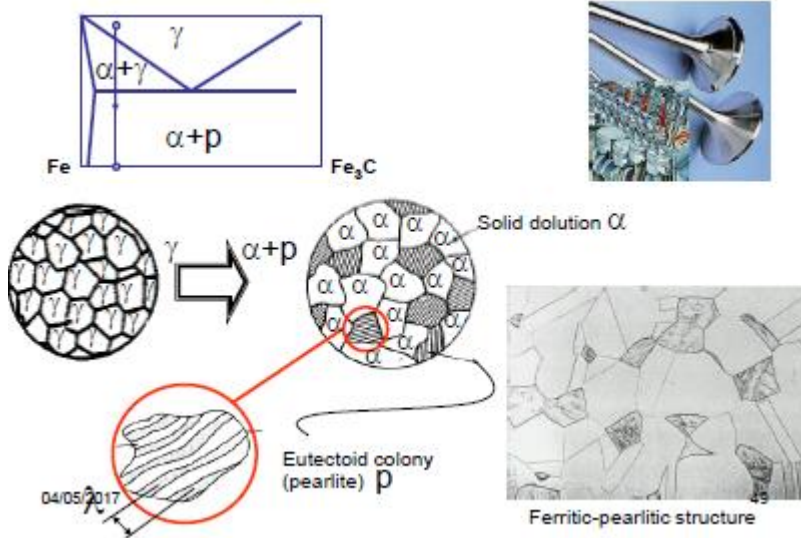
- refining the grain (GB pinning))
- dispersion of particle into the matrix (at GBs)

Strain hardness structure+ GBs particles

We disperse into the liquid some particles already produced (particles are usually very small because large particles cause brittleness). They are usually Silicon carbides and alumina. If the contribution of these particles is quite high compared to the liquid, we have the metal matrix composite (strengthening by adding a new hard alloy). Whenever the liquid will be solidified, these particles will locate at

the gbs of the solid → slight difference between precipitation and dispersion hardening where we have strengthening because we add a volume fraction of hard phase → the larger the number of particles, the larger the strengthening.

6) Strengthening by phase transformation



Strengthening a ductile material by introducing a hard phase promoted by solid state heat treatment.

There are alloys which have a polymorphic behavior (like Fe, which can change its crystal structure from room T to high T ferrite, austenite, ... produced by changing T)

Not only steels, but also Titanium has polymorphic behavior.

We can exploit heat treatment by phase change to promote new mechanical properties of the alloys → by controlling cooling of initial microstructure from γ to room T, we

will produce again ferrite and pearlite according to phase diagram.

At faster cooling with increasing cooling rate, pearlite will become finer, closer λ (lamellar distance). This second phase will contribute to the increase of mechanical strength of steels very dramatically.

According to previous strengthening mechanism by precipitation, we have also steels that can be hardened at very significant level to obtain high strength and the strength in this case can achieve up to 2 GPa → steels strengthened by precipitation.

Steels are stainless: anti-corrosive, the highest strength in steels (stainless steel → a form of steel containing chromium, resistant to tarnishing and rust).

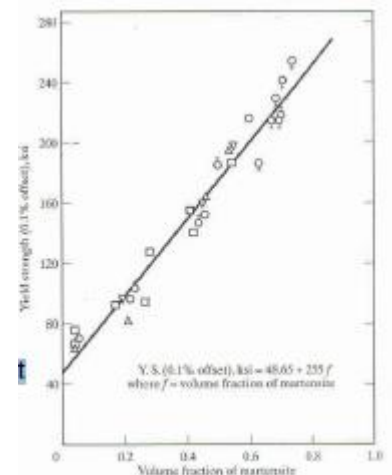
Strengthening by heat treatment



Ferritic structure (light gray) with distributed martensite (dark)

Strengthening of a stainless steel (304) due to martensite microconstituent (distributed hard phase): austenite is transformed into martensite after heat treatment and plastic deformation.

In general, all polymorphic alloys could exploit strengthening by heat treatment (es. Titanium alloys) with the presence of a distributed hard phase.



7) Strengthening by fibers

The formation of fiber features into a metallic matrix will enhance the mechanical properties of the matrix, even up to 5-10 fold, due to the strengthening action induced by fibers

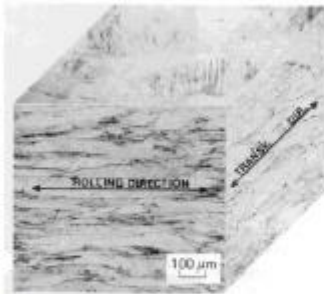
Fibers operate as obstacles to:

- Crack propagation into the matrix (toughening)
- Dislocation motion (strengthening)

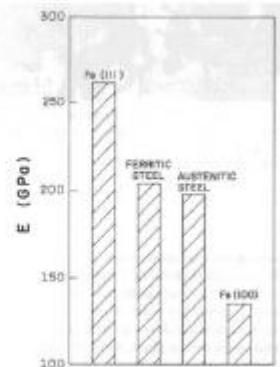
When we have a manufacturing process like rolling, the material becomes thinner by two rolls. Another outcome of rolling is to give the material a texture, fibers → mechanical properties along the fibers will be larger than the transversal direction. That's not always convenient because by rolling we inevitably produce anisotropic properties → the only way to make the material homogenous is to homogenize everything by annealing which reduces the strength we got from rolling.

Fibers are also common in metal matrix composites. These fibers are not produced by processing, but they're added into the melt. After solidification, the fibers will strengthen the alloy.

8) Strengthening by texture



Ni-200 cold worked by 60% thickness reduction

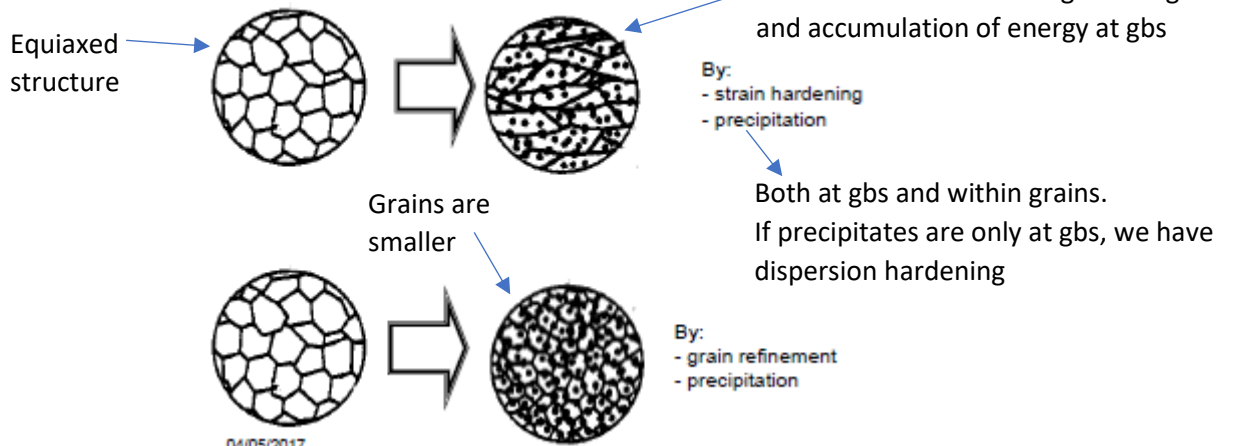


Elastic modulus E limits for steels

Grains elongated along the rolling direction. Black regions → plastic inclusions because they can elongate along direction of rolling and they have a strengthening mechanism like fibers.

Energy absorbed and toughness are higher along rolling direction.

Complex Strengthening methods: schematics



Complex Strengthening methods: AHSS steels

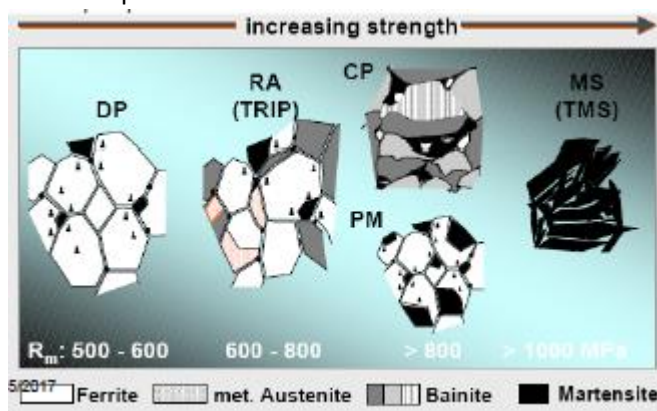
Multiphase steels used for deep drawing operations (automotive): hot rolled + controlled quench cooling

DP = Dual Phase

TRIP = Transformation Induced Plasticity

MS = Martensite

CP = Complex Phase



AHSS: Advanced High Strength Steels, 3rd generation of advanced steels. They are characterized by high formability (shaping and high strength so material is less prone to dendrites). Once we produce the alloy by making martensite originate into ferrite matrix, we get dislocations. Presence of dislocations means there is strain hardening, they are around the black spots only. They are strictly related to martensite, which is produced by expansion during heat treatment: $\gamma \rightarrow \alpha'$ (martensite) produced with increase of

volume (expansion). Martensite will press neighboring grains by introducing additional dislocations.

Dislocations: cause of the "knee" in the strain-stress curve. As strain increases, dislocation will accumulate starting from that region; then the material will expand in yielding into grains.

Consequence of strengthening: austenite can be transformed into martensite by increasing strength → high dendrite resistance; transformation by cooling and by plastic deformation. When austenite is in metastable condition, it can be transformed. Also heat treatment and electromagnetic energy can transform it → TRIP. Austenite has FCC structure → it can be plastically deformed.

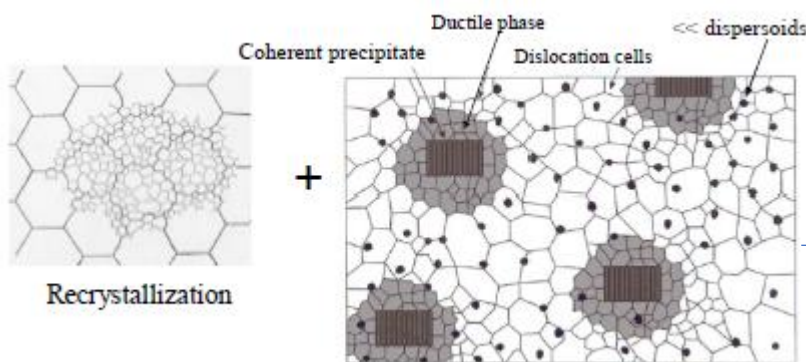
CP → very hard to identify the phases as Bainite has three possibilities of grading.

PM (Partial Martensite) → amount of ferrite lower compared to DP; less plasticity, PM used whenever we don't need to shape in a complex geometry.

Martensite hardness depends on amount of Carbon which is very small because we need formability <0,2%; Martensite we use is strong as microstructure not as hardness (soft martensite with less C).

Strengthening in nanostructures

Plastic deformation + recrystallization + precipitation/controlled dispersion



This kind of microstructure is the most advanced principle to make alloys strong and tough → we have small regions where we have precipitates which are prismatic (half coherent)

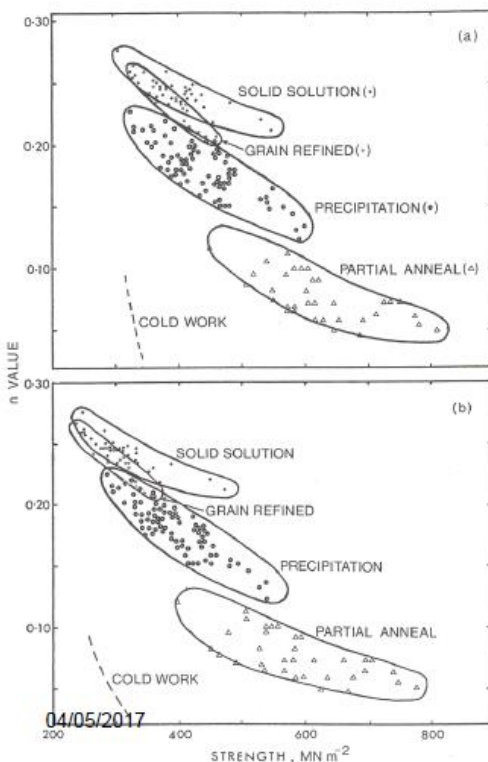
→ Damascus swords → we have small, hard, round particles, surrounded by a soft matrix → fineness of microstructure gives us toughness

(result of impact energy absorption) → high energy absorption when any crack formed at interface between particles and soft matrix will stop growing during stressing of material.

Any hard phase into soft phase → at interphase, we should have crack which will stop growing during stressing of material if around precipitates there's some soft phase.

The smaller the grains (fine microstructure), the more difficult for the crack to propagate.

Effect of strengthening on strain hardening coefficient n



We need to optimize strength and formability → important to know which of the strengthening mechanisms can make favorable plastic deformation shaping.

← Comparison between strength and n → highest n, highest formability

Banana – like curves constructed from each strengthening mechanism. Data collected from tensile test

Solid solution is the best strengthening mechanism that can allow highest n. The presence of even small amount of solid solution → distortion → when we make plastic deformation, dislocations will interact with this distortion to strengthen the material.

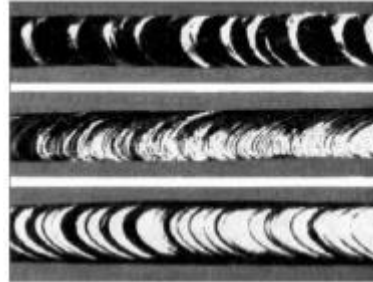
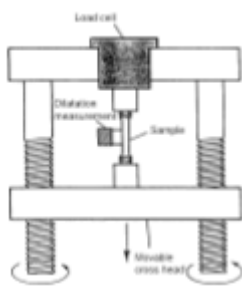
Solid solution: interstitial (better formability, very dramatic, small size fraction) and substitutional (small size factor but not very dramatic) → strengthening in homogeneous way. Even small amount makes very dramatic distortion and dislocations will be trapped into those regions.

To make solid solution, we need slightly 50% difference in diameter which will not make very high distortion even though it will make strengthening.

Order of strengthening methods from best to worst:

1. Solid solution
2. Grain refinement
3. Precipitation
4. Partial annealing → we have strengthened the alloy and slowly heated it up → Largest strength retained
5. Cold work → worst, because we have many dislocations

Tensile test of HCP monocrystal metals



Macroscopic effect of dislocation motions (slip bands) in hexagonal monocrystals [Kleber 1990]

In monocrystals, slip occurs along the sample length by a number of equivalent favorable sliding planes and directions; The plastic deformation is shown by visible parallel ring-shaped superficial steps

towards the periphery of the sample.

Each step results from the reciprocal collapse of a large number of dislocations along the same sliding plane. After surface polishing these rings appear as thick lines.

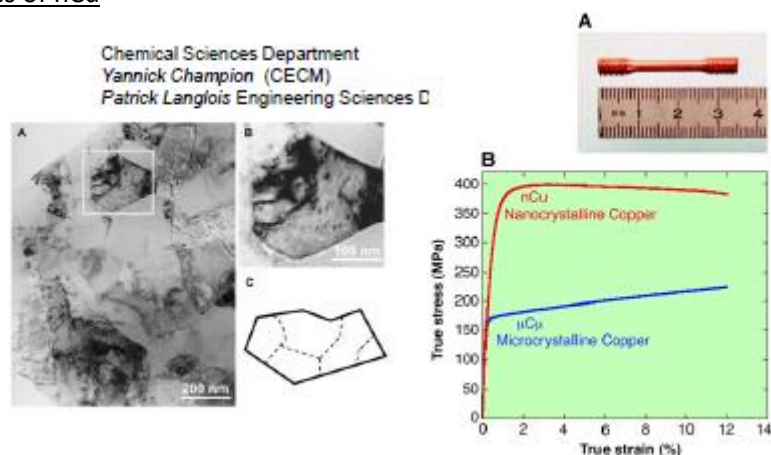
Ideal microstructure for “strength”?

- Can't use a **non-metal** as base of alloy
 - could get high yield strength this way but toughness would be far too low.
- Can use **work-hardening** to improve strength, but:
 - effectiveness very limited for materials with yield stress increased close to UTS by other means
 - reduces toughness (by reducing work-hardening capacity)
 - will “anneal out” at moderate temperatures
- Can use **fine grain size** to improve strength /and toughness!)
 - relatively limited (but useful) strength improvement (a few hundred MPa)
 - fine grain size is no good at high temperatures because the grain will grow and, since grain boundaries are fast diffusion paths, creep strength will be poor.
- Can use **solution strengthening**
 - in BCC, interstitial solutes give a limited but useful increase in strength [+50 to 80 MPa]
 - (most steels are well over the solubility limit for C anyway)
 - can quench in excess C N into solution in martensite (up to +1000 MPa)
 - but this form of martensite is very brittle (all dislocations are locked)
 - substitutional solutes can give moderate strength increases (a few hundred MPa)
 - limited by solubility – solutes with big misfits will have low solubilities
- Can use **precipitation hardening**
 - need to achieve a very fine dispersion of very hard precipitates (up to +1000 MPa)
 - or a very fine dispersion of weaker precipitates with strong associated strain fields (not quite so effective)

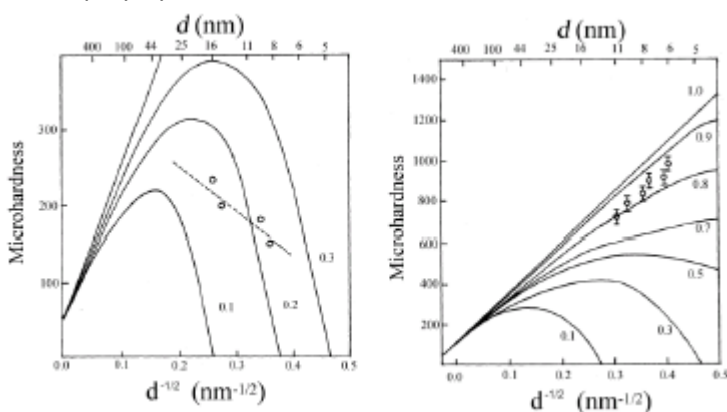
"Ideal" Microstructures

- Use **precipitation hardening**
 - need as fine a dispersion as possible of particles as hard as possible.
- Topped up with **solution strengthening**
 - Solutes may be present anyway for a variety of reasons:
 - to scavenge undesirable impurities (e.g. Mn, Si, Al in steels)
 - to slow down diffusional reactions and thus promote martensite
 - to confer corrosion resistance (e.g. Cr in steels)
 - to react with other elements present and produce precipitates
- Why not have a **fine grain size** anyway?
 - Generally a good idea, as toughness is improved
- (Can we make spatial variations: e.g. surface different from bulk?)
- Extra constraints if **high temperature** use is likely:
 - base material must have a high melting point (!)
 - any precipitates must be "stable" (there is a dynamic driving force to reduce number and increase size)
 - reduce driving force (interface energy)
 - slow down "coarsening" (reduce diffusion rate)
 - fine grain size may not be good
 - grain boundaries are fast diffusion paths & can get high creep rates
 - grain size will tend to grow anyway
 - beware (or use!) further reactions between elements present.

Mechanical properties of nCu



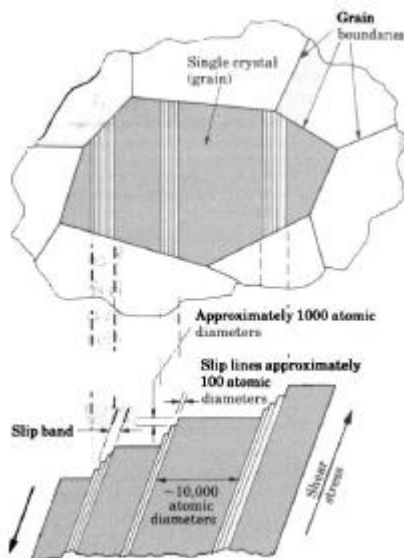
GBs in polycrystalline Cu, Fe



Microhardness as a function of grain size for different GB densities down to Nanocrystalline size:

- a) Cu
- b) Fe

Slip lines and slip bands

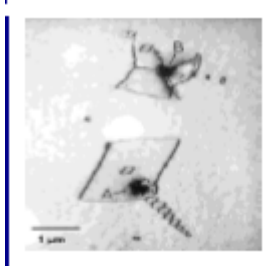


Slip lines and bands in a stressed monocrystal. A slipband is composed of many slip planes. Upper figure shows a grain surrounded by other grains. The bottom figure shows the difference between slip lines and slip bands.

Precipitates and Dislocations

Precipitates usually do not fit into the host lattice. The growing particle causes considerable stress that can be reduced by plastic deformation.

If the precipitate fits in one lattice direction, but not in others (a precipitate with a hexagonal lattice, e.g., may fit relatively well on the $\{111\}$ planes of an FCC lattice) a compromise between a non-spherical shape of the precipitate and a system of dislocation loops in some direction may produce least strain energy. The precipitate-dislocation system then has a very specific structure; the process is known as "prismatic loop punching". An example is shown below on the left (taken under kinematic bright field conditions).



Precipitate with prismatic loops.
An arrangement like that
accounts for the peculiar etch

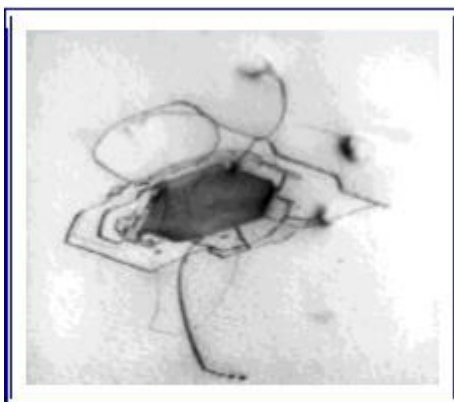


Plate-like precipitate (the dark grey feature) with dislocations
relieving parts of the stress.

Yielding in Polycrystalline Materials

- Polycrystals: in the absence of texture can be treated as isotropic materials
- Dislocations movement is hindered because is restricted to one grain
- When a polycrystalline material is subject to an imposed stress above its yield stress, slip occurs in the material as movement of dislocations along the slip plane ($\tau_{RSS} > \tau_{CRSS}$), this is a necessary and sufficient condition for slip system activation.
- Deformation on only one slip system is not possible because various grains have to be compatible
 - Deformation is inherently inhomogeneous (it varies from grain to grain)
- When a crystal is surrounded by other crystals of different crystallographic orientation, deformation of the crystal cannot start at the primary system, as the strain taking place need to be compatible at the boundary with the strain in the other crystals (not discontinuities along the grain boundary).

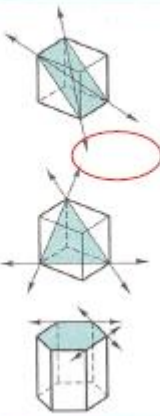
Microplasticity in polycrystals

- For slip to occur in polycrystalline materials (ductile behavior), at least 5 independent slip systems are required.
- Materials which are ductile in single crystalline form, may not be ductile in polycrystalline form.
- FCC crystals (Cu, Al, Au) have excellent ductility.
- At higher temperatures more slip systems may become active and hence polycrystalline materials which are brittle at low temperature, may become ductile at high temperature.

Lecture 7

Plastic deformation in polycrystals

Slip systems in cubic crystals

Crystal structure	Slip plane	Slip direction	Number of slip systems	Unit cell geometry	Examples
bcc	{110}	$\langle \bar{1}11 \rangle$	$6 \times 2 = 12$		α -Fe, Mo, W
	{112}		12		
	{123}		24		
			In totale: 48		
fcc	{111}	$\langle 1\bar{1}0 \rangle$	$4 \times 3 = 12$		Al, Cu, γ -Fe, Ni
hcp	(0001)	$\langle 11\bar{2}0 \rangle$	$1 \times 3 = 3$		Cd, Mg, α -Ti, Zn

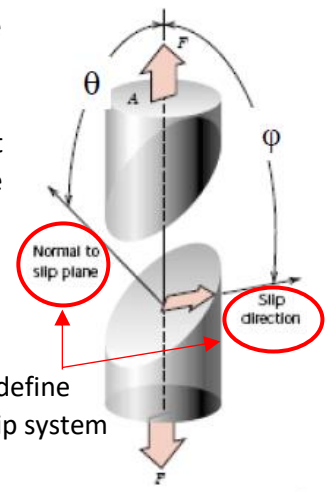
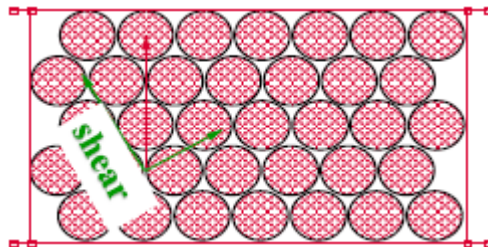
Dislocations start moving on slip planes along slip directions. The slip planes are the densest packed and the slip direction are the ones of greatest density.

The slip direction must lay on the slip plane in order to have a slip system. We geometrically verify this condition by the dot product of the two (it is equal to 0 when the direction lays on the plane).

Plastic Deformation under normal stress

In order for the dislocation to interact with other defects to strengthen the material, it must move on slip planes; if we have a shear stress to push dislocations, that's acceptable. We can also apply a normal stress.

Any normal force F can be resolved into a shear τ and a normal σ stress component over a slip plane. When the resolved shear stress τ , exceeds some critical value τ_{CRSS} , slip occurs between adjacent "slip planes".



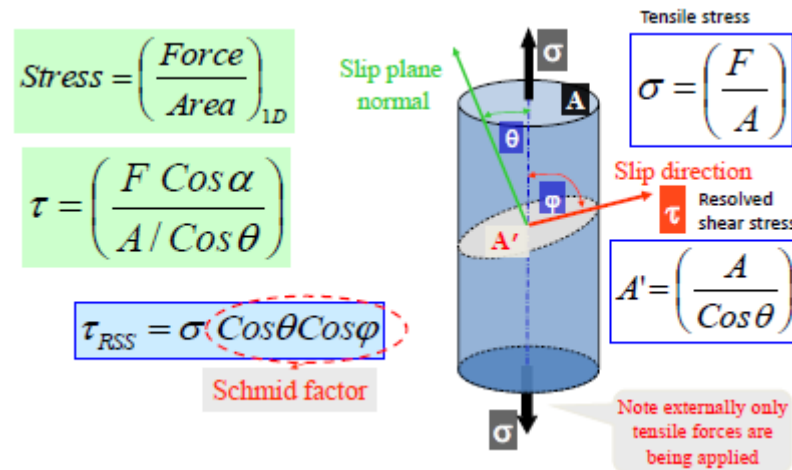
$\sigma \perp \text{surface}$ while $\tau \parallel \text{surface}$

Decomposition done just by mathematical mean $\rightarrow$ we may define the angles (θ and ϕ) $\rightarrow$ slip system defined with respect to the applied force.

The Schmid's Law

- In order for a dislocation to move over its slip system, a shear stress acting over the slip plane and along the slip direction must rise resulting from the applied normal force.
- A uniform tensile stress σ originates a **resolved shear stress** on planes that lie at an angle θ to the tensile axis; It is just this resolved shear stress that is responsible of dislocations motion, when its magnitude exceeds a critical level τ_{CRSS} ,
- **Critical Resolved Shear Stress** τ_{CRSS} is the stress at which macroscopic yield begins $\rightarrow$ dislocations start moving.
- Each material has its own τ_{CRSS} , so it is a material property.
- It is closely related to yielding stress σ_y .
- When a tensile stress is imposed to a metal, dislocations will first move on the slip plane on which τ_{CRSS} is minimum.

Mathematical Derivation of the Schmid's Law



Critical Resolved Shear Stress

If the slip plane is perpendicular to the Tensile Stress $\theta = 90^\circ$, and $\phi = 0^\circ$

$$\cos \phi = 1 \text{ and } \cos \theta = 0 \Rightarrow \tau_{RSS} = 0$$

This means that even if the applied tensile stress is enormous, the resolved shear stress is simply zero. The dislocations can't move (slip can't occur).

For these extreme orientations, the crystal ordinarily fractures rather than deforming plastically.
 $\Rightarrow$ maximized efficiency of applied tensile stress with respect to angles orientation

To sum up, to promote the movement of dislocations, we don't necessarily need the shear stress. we can apply normal stress and only the component projected over the slip plane and slip direction will act for plastic flow.

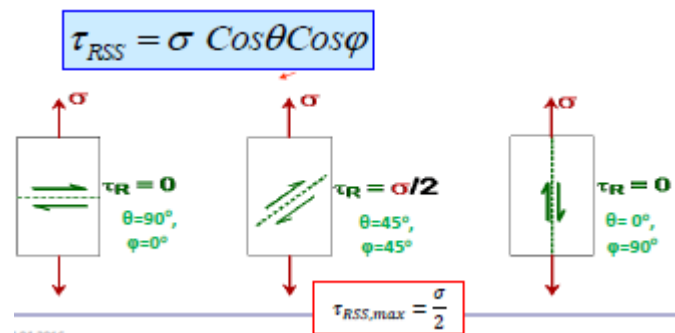
Critical Resolved Shear Stress

Condition for dislocation motion: $\tau_{RSS} > \tau_{CRSS}$

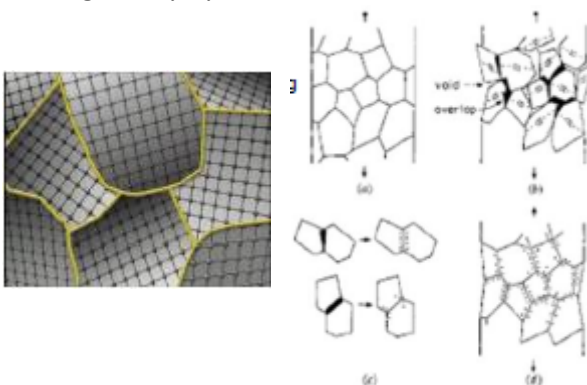
Typically $10^{-4}G$ to $10^{-2}G$

Crystal orientation with respect to imposed σ can make it easy or hard to move dislocation.

Very small shear stress can be sufficient for dislocations to move.



Yielding in Polycrystalline Materials



Strain is transmitted from grain to grain whereas the grain assembly remains cohesive (i.e. no voids or cracks form inside the material).

Eventually, if the load is sufficiently high, damage processes involving crack nucleation and growth may set in and eventually result in the ultimate failure of the polycrystalline material.

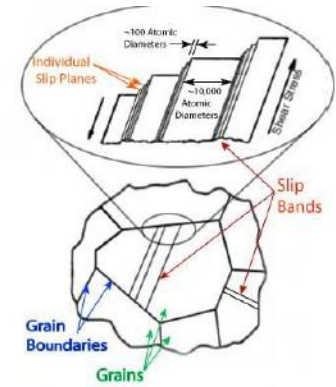
Many single crystals with random orientation and any grain is defined by its GB $\rightarrow$ no voids and no overlaps. GBs connected to each other by some cohesive energy.

Plastic Deformation in Polycrystals

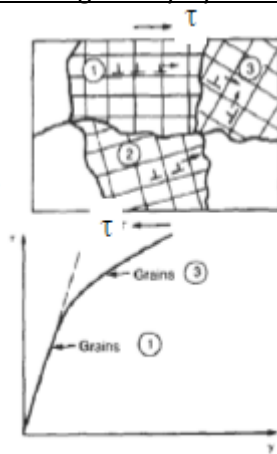
- Slip planes & directions (θ, φ) of the same slip system change from one grain to another,
- Resolved Shear Stress will vary from one grain to another $\rightarrow$ different Schmid's factor from one grain to another, which leads to different slip directions.
- The grain with the largest Resolved Shear Stress (slip planes as nearly parallel to τ as possible) yields first. Other (less favorably oriented) grains yield later.

Once the grains start flowing and get saturated (many dislocations will nucleate in this plane and they will hinder to each other by slowing down), no dislocation can move and the stress applied will activate another grain $\rightarrow$ the most favorable after the first one.

Grains will never flow simultaneous, they will strain one by one, according to their orientation with respect to τ .



Yielding in Polycrystalline Materials



← Slip planes as nearly parallel to τ as possible $\rightarrow$ grain (1) will slip first.

- Slip begins in grains where there are slip planes as nearly parallel to τ as possible, e.g. grain (1). After this grain saturates slip spreads to grains like (2) which is not so favorably oriented, and lastly to the worst oriented grains like (3).
- Yielding does not take place all at once, therefore, and there is no sharp polycrystalline yield point on the stress-strain curve.

Minimum # of slip systems for homogeneous strain in crystals

For a polycrystalline material to have appreciable ductility, each of its grains must be able to undergo the same shape change as the entire body. That is, each grain in a polycrystal must deform with the same external strains as the whole polycrystal $\rightarrow$ **compatibility constraint** (when we make plastic deformation, the final strain in an overall crystal must be compatible with the strain of any crystal grains)

For an individual grain, this amounts to an imposed set of strains along the crystal axes

$$\epsilon_1, \epsilon_2, \gamma_{23}, \gamma_{31} \text{ and } \gamma_{12}$$

The number of independent slip systems is equal to the number of strain components that can be accommodated by slip.

The third normal strain is not independent because: $\epsilon_3 = -\epsilon_1 - \epsilon_2$

If a material has less than five independent slip systems, a polycrystal of that material will have very limited ductility (to make plastic deformation possible) unless other deformation mechanism (e.g. twinning) supplies the added freedom necessary.

Schmid's Law for polycrystals

It can be shown that in 2D crystals a minimum of 2 slip systems and in the 3D case, a minimum of 5 slip systems are needed to satisfy an arbitrary shape change for ductile materials (provided that no constraints exist)

- A slip system is independent of other if the deformation it produces cannot be replaced by a combination of shears on the other slip systems

FCC crystals have 12 different slip systems $\Rightarrow$ 384 different combinations of 5 independent slip systems which will generate an arbitrary shape change Taylor showed that a set of slip systems is selected for which the total shear

$$\Gamma = \sum_{i=1,5} d\gamma_i = MIN$$

$\Rightarrow$ the orientation change during deformation and the strain hardening behavior of polycrystals to be calculated.

First, the average Schmid's factor has to be determined: in analogy to single crystals:

$$d\varepsilon = m d\gamma$$

In polycrystals: $d\varepsilon = m_T d\Gamma = m_T \sum d\gamma$

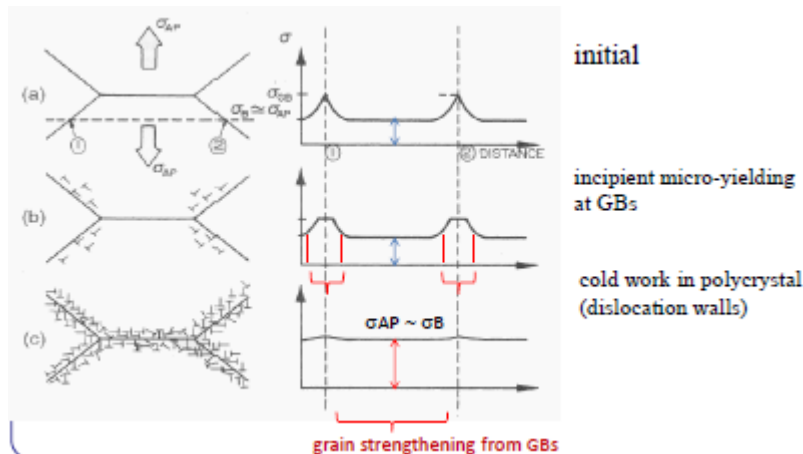
For random orientation distribution, Taylor calculated that $M_T = 1/m_T = 3.06$

The average Schmid factor for polycrystals during tensile deformation is $m_T = 1/3.06 = 0.327$

All the strength applied on the polycrystal is uniformly strained due to random orientation, even though each grain is oriented in a different way $\rightarrow$ no crystal anisotropy.

Plastic deformation in polycrystals

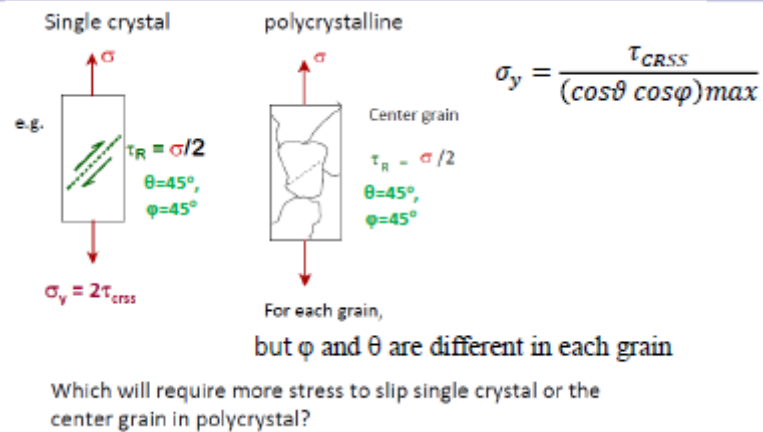
We have two grains in contact with each other, linked by some GB energy. What happens when we apply stress?



Plastic flow not uniform in grain, but it starts from GBs $\rightarrow$ when grains are in contact and normal stress is applied to gbs, the stress has a higher value in points 1 and 2 in GBs. If σ increases, the overall boundary of gbs will collect a number of dislocations. Therefore the area around the initial cross-section $\rightarrow$ the region with dislocations gets larger and larger as σ increases. In the rest of the grain τ will remain the same as "initial" state.

Only when all GBs are saturated by the increase of stress, also the core of the grain will increase in stress $\rightarrow$ σ in Gbs equal to σ in the grain.

The Schmid's law for Polycrystals



Relation between work hardening coefficient and uniform strain

$$P = \sigma_f \cdot A$$

$$dP|_{necking} = A \cdot d\sigma_f + \sigma_f \cdot dA = 0 \quad (1)$$

$$V = A \cdot \ell = \text{const.}$$

$$dV = A \cdot d\ell + \ell \cdot dA = 0 \rightarrow dA = -A \cdot \frac{d\ell}{\ell} = -A \cdot d\varepsilon \quad (2)$$

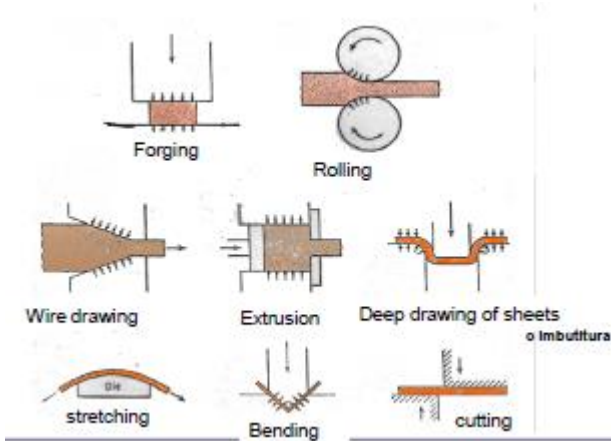
$$\sigma_f \approx K \cdot \varepsilon^n \rightarrow d\sigma_f \approx n \cdot K \cdot \varepsilon^{n-1} \cdot d\varepsilon = \frac{n}{\varepsilon} \cdot \sigma_f \cdot d\varepsilon \quad (3)$$

$$dP|_{necking} = \frac{n}{\varepsilon} \cdot \sigma \cdot d\varepsilon - \sigma \cdot d\varepsilon = 0 \quad \therefore n = \varepsilon_u$$

Lecture 8

Shaping by plastic deformation: hot, cold and warm

Main plastic deformation processes



Annealing: heating the metal and allowing it to cool slowly, in order to remove internal stress and toughen it.

Forging: shaping the metal by heating it and hammering it

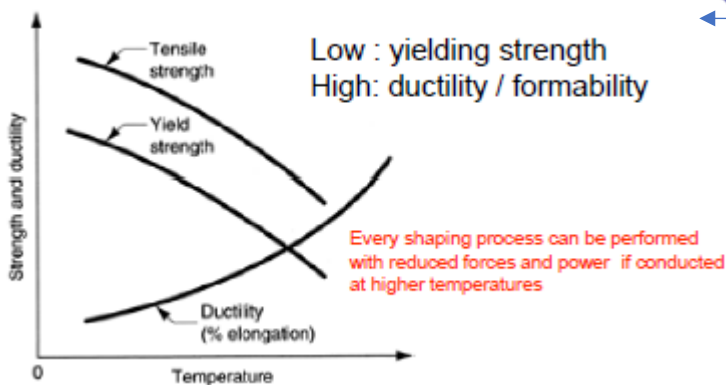
Rolling: the metal stock is passed through a pair of rolls to reduce the thickness and to make the thickness uniform. The rolls are counter rotating

Wire-drawing: drawing out the metal into wire by passing it through a series of holes of diminishing diameter in a steel plate → metal heavily strain hardened.

Extrusion (most of Al alloys): the material is pushed through a die of the desired cross-section

Deep drawing: a sheet metal blank is radically drawn into a forming die by the mechanical action of a punch

High temperature properties

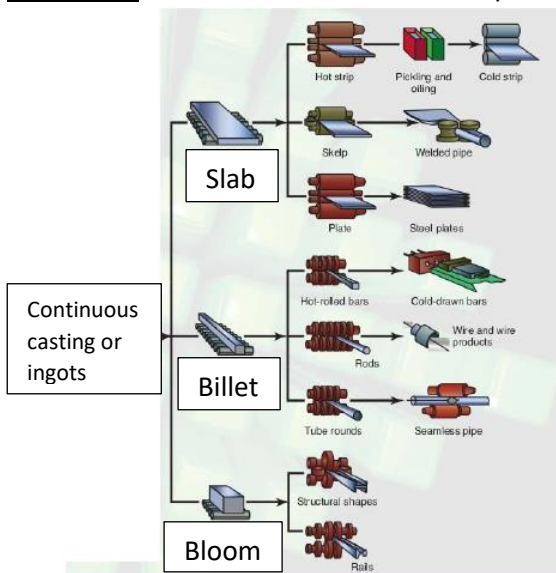


← At what temperature we have to shape our components to have perfect geometry and mechanical properties.

We can get large displacements with increasing temperature and we may use presses with lower forces.

The ductility increases and the yielding stress decreases when the processing temperature increases.

Hot Rolling → we start with different shapes of material depending on their cross section

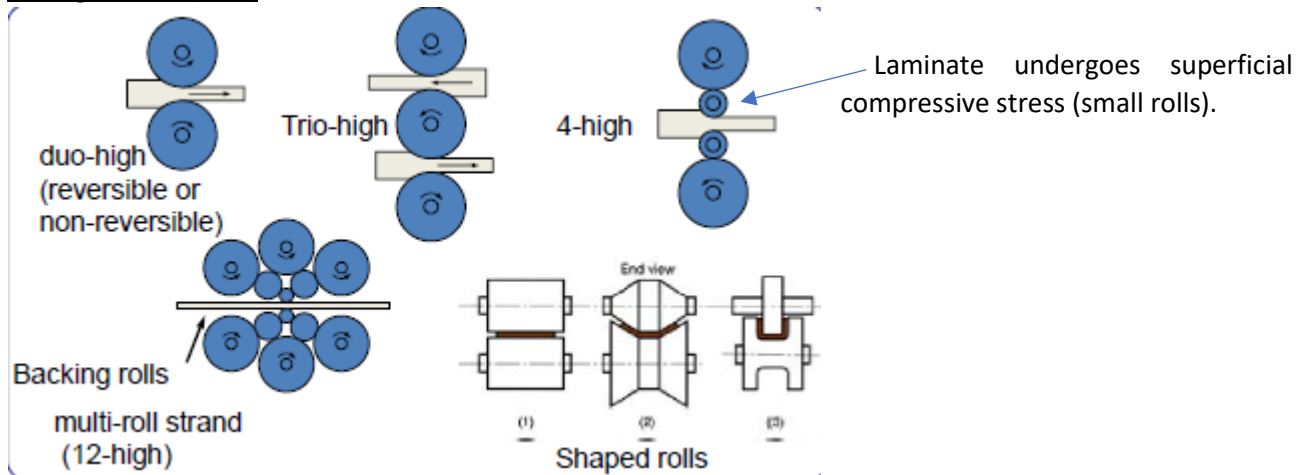


→ Sheets, pipes, plates, ...

→ Bars, rods, wires, round tubes, ...

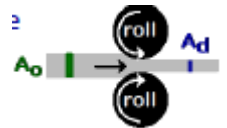
→ Structural shapes, rails, ...

Rolling strands (rolls)



Hot rolling

- Large portions of material (slab or billet) are shaped while it is heated at the recrystallization temperature ($1/3 < T_m < 1/2$)
- One or a sequence of passes is needed in roll shaping
- Rolls exhibit a conveniently shape to produce sheets, wires, long bars of various cross sections
- Hot rolling enables large thickness reduction at the recrystallization temperature of the alloy, much larger than those exerted by cold rolling, keeping constant the number of passes
- Recrystallization ensures the largest ductility of the material and preserves minimum equiaxed grain size

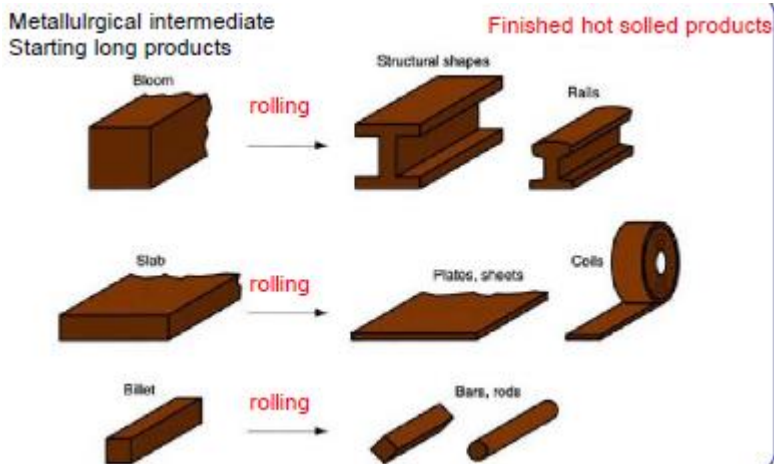


Hot rolling workshop



Rolling strand

Hot-rolled (long) products



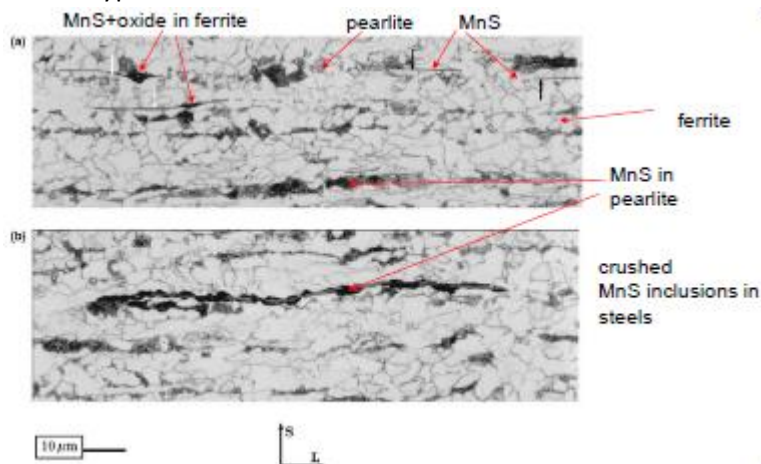
Hot-rolled and cold-rolled sections



Rolling provides the largest variety of sections; in addition, it can shape much larger material volumes compared to other processes.

Hollow sections are very important in many applications because they exhibit higher strength with lower weight since they have a cavity inside.

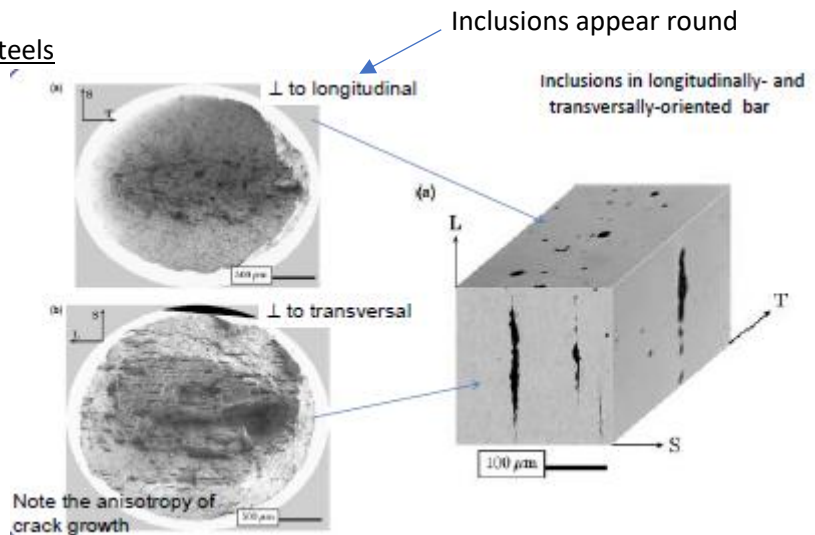
Rolled Hypo-eutectoid Steel Microstructure



Texture during rolling is important. The plastic deformation of manganese sulfide occurs along the rolling direction in which material becomes stronger.

Since the solubility of sulfur in iron and steel is very low, it is usually present as sulfide. The sulfide inclusion formed during solidification is predominantly manganese sulfide, MnS: high T_m ; improves machinability and retards grain growth.

MnS inclusions in steels

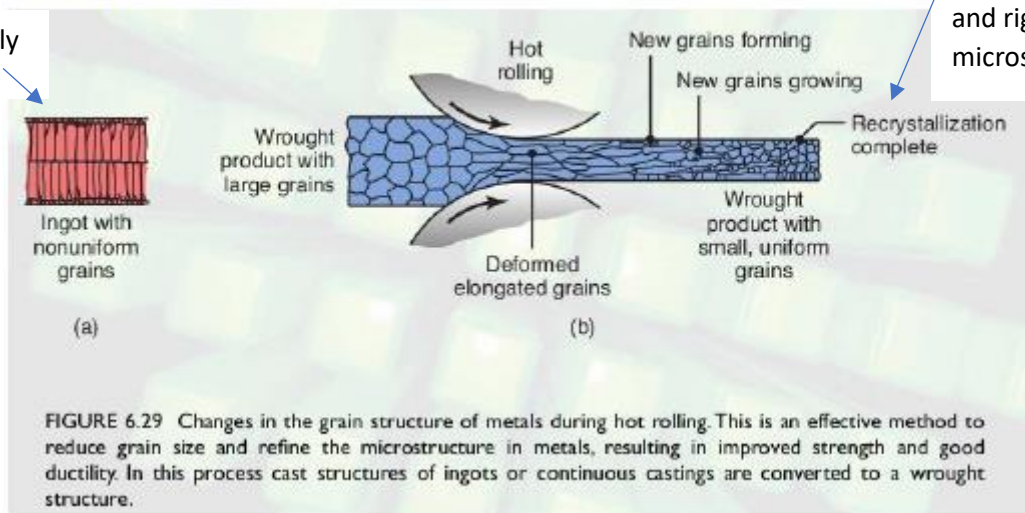


If we have a volume extracted by a laminate, we can see the rolling directions (longitudinal and transversal).

Metallurgical processes during hot rolling

Recrystallization

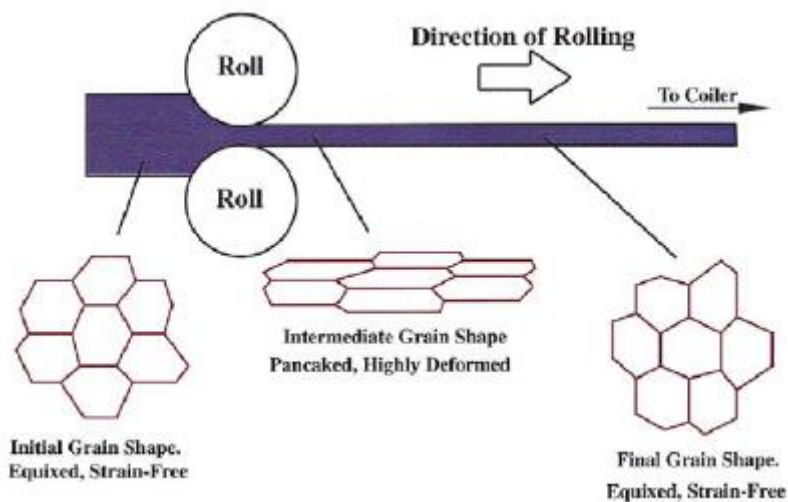
Ingot initially solidified



Right thickness and right microstructure

FIGURE 6.29 Changes in the grain structure of metals during hot rolling. This is an effective method to reduce grain size and refine the microstructure in metals, resulting in improved strength and good ductility. In this process cast structures of ingots or continuous castings are converted to a wrought structure.

Because of the high energy accumulated at GBs and because of temperature (recrystallization), new grains are formed during rolling → recrystallization has occurred.



Recrystallization phenomena vs. Time/Temperature

Slip bands → collections of dislocations



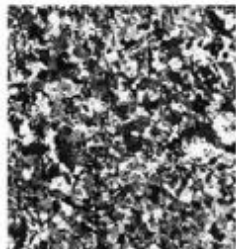
Cold-worked (33%CW)



Initial stage of rec. after heating 3 s at 580°C



Partial replacement of cw grains by rec. ones 4 s.



Complete recrystallization (8 s at 580°C).



Grain growth after 15 min at 580°C



Grain growth after 10 min at 700°C

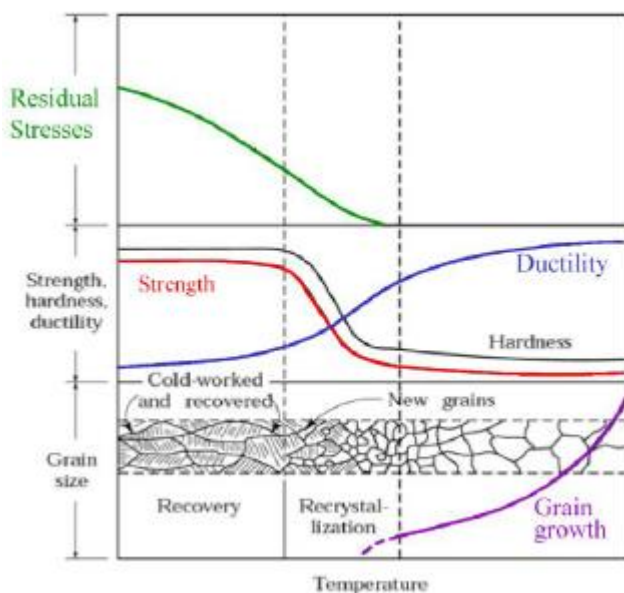
When we heat the alloy at recrystallization temperature, new grains start forming (small grains). The higher the temperature, the longer the time → time and temperature work in the same way; same results.

If we want a very fine microstructure, we freeze this microstructure at this point rapidly by quenching → the alloy is stress and strain free → if we apply a force to reduce the thickness, we can achieve the largest thickness reduction with minimum applied force.

Grains have twinning inside → we allow relaxation after deformation to see

it

Materials properties changes upon hot rolling



At lower temperature, the material will undergo stress relaxation, we are not losing properties (such as hardness, strength, ...) → we maximize the compromise between toughness and strength → high strength and improved toughness.

In the recrystallization region, residual stress is completely eliminated; strength decreases quite remarkably; ductility increases; microstructure → finer grains nucleated and started to grow.

At higher temperature and longer time, grains continue to grow → one single grain (cannibalism)

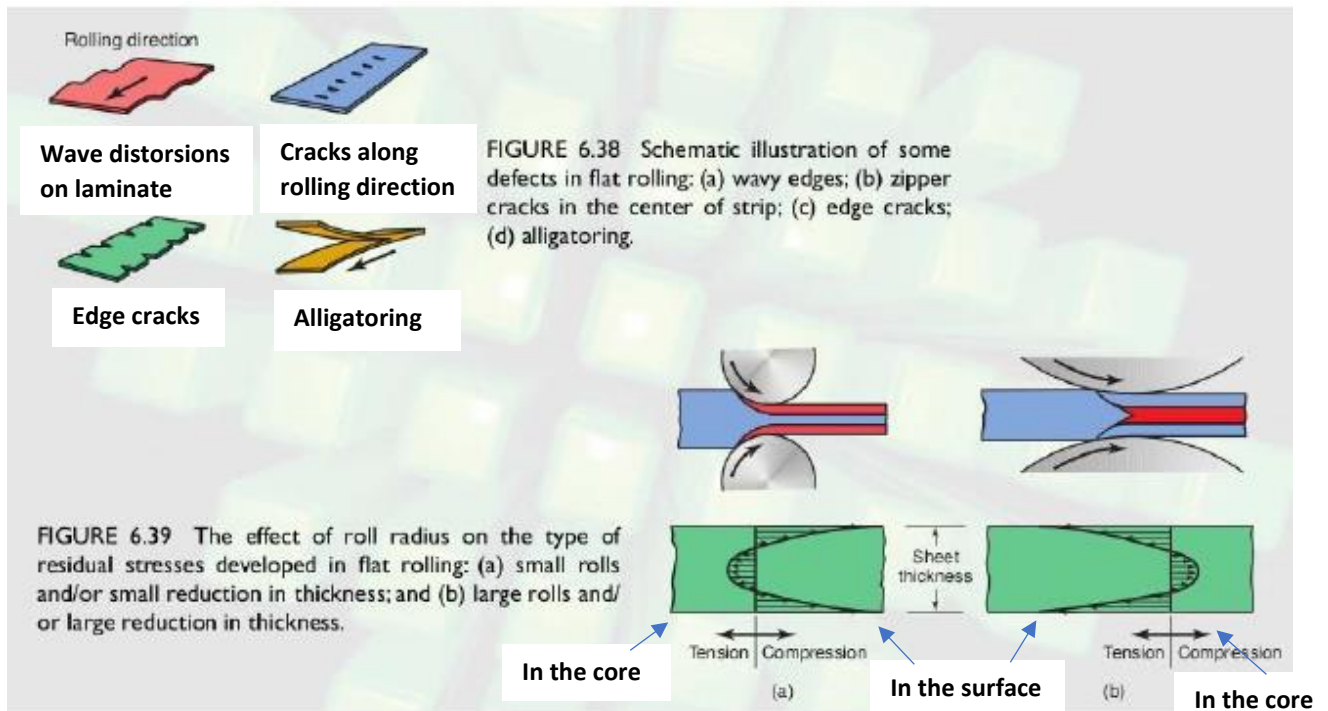
Cold rolling

It occurs with metal below its recrystallization temperature. It is characterized by:

- Improved surface finish
- Larger force required compared to hot rolling (the material is quite strong and difficult to deform)
- To enhance further surface finish and dimensional control
- More complex roll systems are required



Rolling defects and induced stress

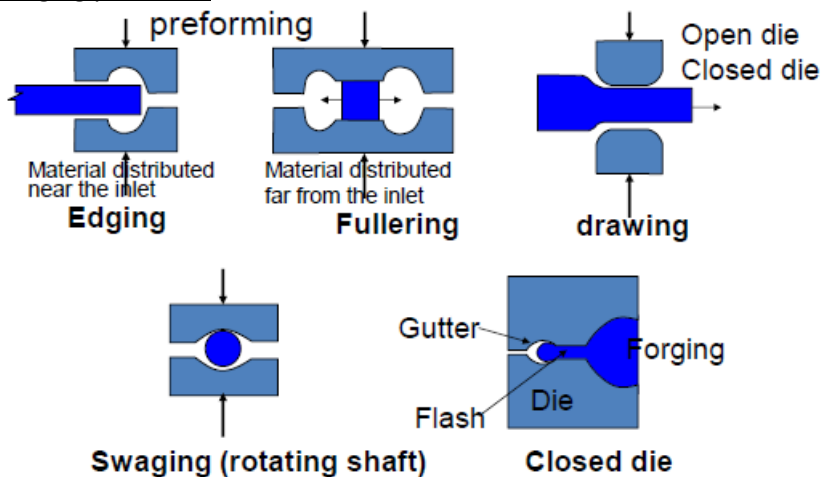


The effect of roll radius on the type of residual stress developed in flat rolling

- (a) small rolls and/or small reduction in the thickness
- (b) large roll and/or a large reduction in the thickness

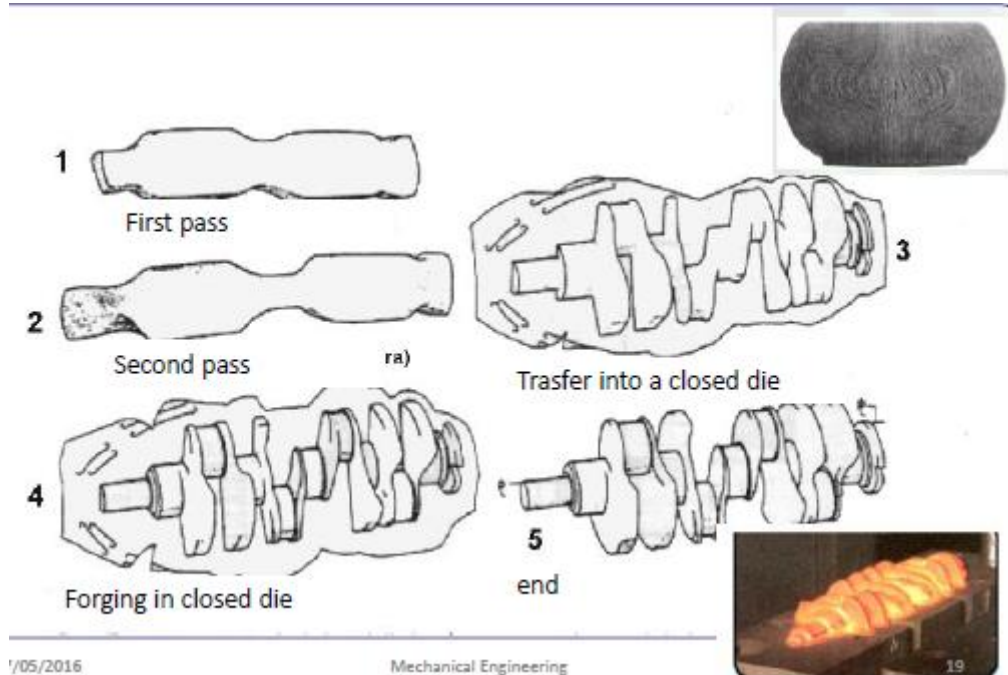
If we have a potential crack, it can be open (it will propagate deeper in the material). If we have a tension on the surface, it will open the crack; if we have compression, the crack will not be easily opened, it keeps it closed → (a) is better! → small rolls in order to impress the compressive stress state (skin-pass process)

Forging processes



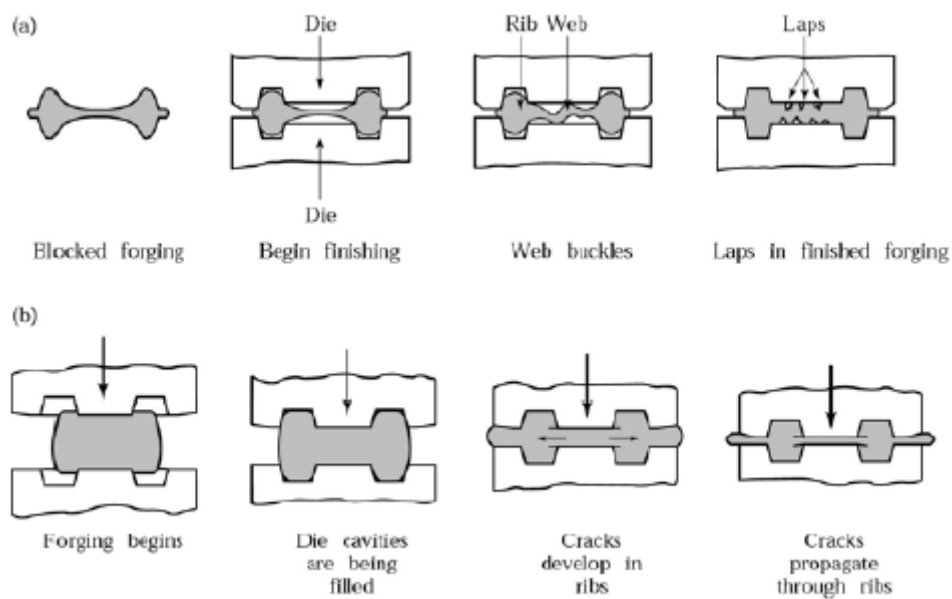
These processes are performed at high temperature because high forces are required to have plastic deformations → the material must be softened for it to be shaped.

Forging with a closed die



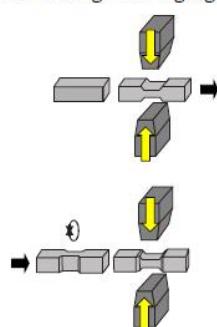
The material is stronger along the direction of fibers and weaker in the transversal direction

Forging defects

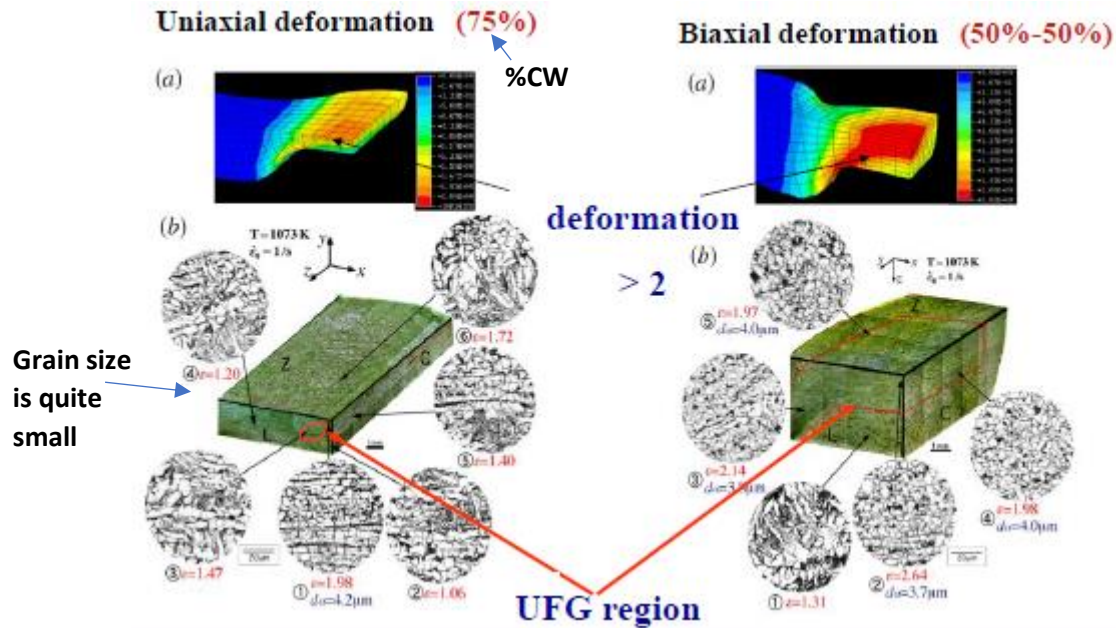


Multi-directional forging to produce nanostructured materials

Laboratory Simulator:
25t rolling & Forging

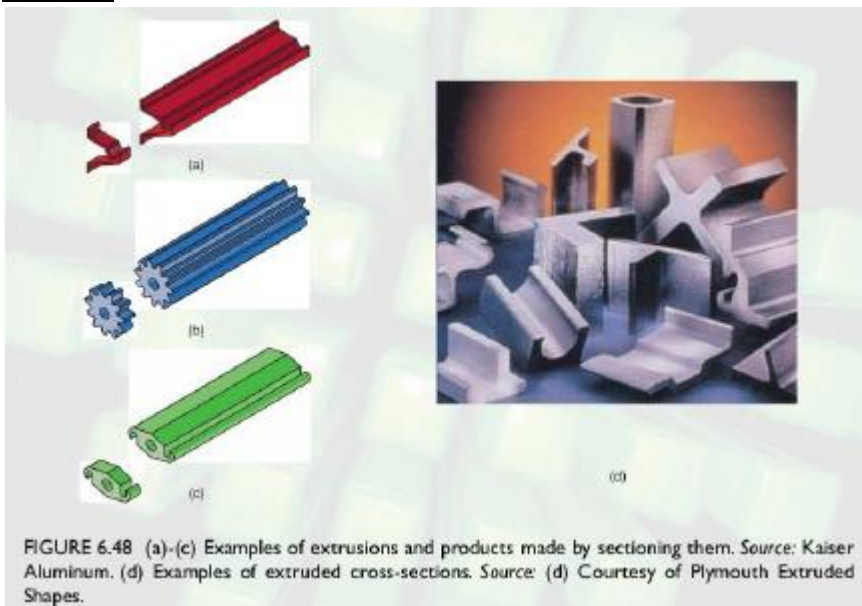


Increasing fraction having UFG microstructure (after plastic deformation)



UFG: Ultra Fine Grain → advanced steels for structural application

Extrusion



Extrusion is mainly used for Al alloys.

The result is long product with a complicated profile.

The material has high resistance so extrusion occurs at high temperature. Al may be initially worked at high temperature and then cold extrusion is applied → high strength, tolerances and good surface finish.

the long product will very complicated profile

Extrusion process types

